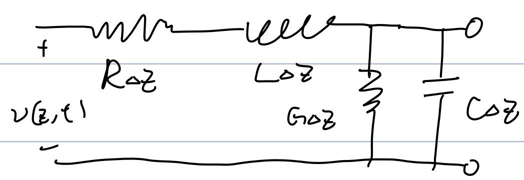

$$\frac{\partial V}{\partial z} = -(R + j\omega L)\, I$$

$$\frac{\partial I}{\partial z} = -(G + j\omega C)\, V$$

$$\Rightarrow \begin{cases} \dfrac{\partial^2 V}{\partial z^2} = -(R + j\omega L)\dfrac{\partial I}{\partial z} \\ \dfrac{\partial^2 I}{\partial z^2} = -(G + j\omega C)\dfrac{\partial V}{\partial z} \end{cases}$$

$$\Rightarrow \begin{cases} \dfrac{\partial^2 V}{\partial z^2} = (R + j\omega L)\cdot(G + j\omega C)\, V = \gamma^2 V \\ \dfrac{\partial^2 I}{\partial z^2} = (R + j\omega L)(G + j\omega C)\, I = \gamma^2 I \end{cases}$$

$$\Rightarrow \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\begin{cases} V = V_0^+ \cdot e^{-\gamma z} + V_0^- \cdot e^{\gamma z} \\ I = I_0^+ \cdot e^{-\gamma z} - I_0^- \cdot e^{\gamma z} \end{cases}$$

$$\frac{\partial V}{\partial z} = -\gamma V_0^+ \cdot e^{-\gamma z} + \gamma V_0^- e^{\gamma z} = -(R + j\omega L)(I_0^+ \cdot e^{-\gamma z} + I_0^- e^{\gamma z})$$

$$\Rightarrow Z_0 = \frac{V_0^+}{I_0^+} = \frac{V_0^-}{I_0^-} = \frac{R + j\omega L}{\gamma} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Set $z = 0$, then

$$Z_L = \frac{V(0)}{I(0)} = \frac{V_0^+ + V_0^-}{I_0^+ - I_0^-} = \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} Z_0$$

$$\Rightarrow \Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$